Prove the number of Hamiltonian cycles of $K_{n,n}$ for $n \geq 2$ is $\frac{n!(n-1)!}{2}$

Asked 6 years, 6 months ago Modified 2 years, 6 months ago Viewed 2k times

1

In wiki , I find the number of Hamiltonian cycles of $K_{n,n}$ for $n \geq 2$ is $\frac{n!(n-1)!}{2}$

Is it true and how to prove it ?

graph-theory hamiltonian-path

Share Cite Edit Follow Flag edited Apr 6, 2016 at 16:34

asked Mar 26, 2016 at 8:02
Lin
167 7

There's a proof given on the page you have attached. Do you need help to understand why it works? Because it's definitely true. – Sarvesh Ravichandran Iyer Mar 26, 2016 at 8:11

1 Answer

Sorted by: Highest score (default)

2

Indeed it's true. Let the bipartitions be A and B, and without loss of generality let us start our cycle at a vertex $a_1 \in A$. Our next vertex must be in B, and since we are working with the complete bipartite graph, we are free to choose any of the n vertices of B, say b_1. Now for our third vertex, we return to A, and we can choose any vertex of A *except* a_1 which we've already used, which gives $n-1$ choices.

You can see that by continuing this pattern, we will have a sequence of vertices

$$a_1,\ b_1,\ a_2,\ b_2, a_3,\ b_3,\ \cdots,$$

eventually returning to a_1 to form a Hamiltonian cycle. At each vertex a_k, for $k > 1$, we have $n-k+1$ choices because we must exclude the previous $k-1$ vertices of A which we've used already. Likewise, at each b_k, for $k \geq 1$ (note that we fixed a_1 but not b_1, so we count $k = 1$ here, whereas we do not for the A vertices), we have $n-k+1$ choices as well.

Multiplying all these choices together, we get a total of $n!(n-1)!$ *oriented* cycles. The factor of $(n-1)!$ comes from our choices of A vertices, and the factor of $n!$ comes from our choices of B vertices. But the cycle we get by choosing

$$a_1,\ b_1,\ a_2,\ b_2,\ a_3,\ b_3,\ \cdots,$$

is the exact same as the cycle we get by choosing

$$a_1,\ b_n,\ a_n,\ b_{n-1},\ a_{n-1},\ b_{n-2},\ a_{n-2},\cdots,$$

just that we are choosing the vertices in the opposite order. Therefore we must divide by 2 to take care of the double counting of cycles due to orientation. This gives the final count at $n!(n-1)!/2$, precisely what you get.

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edited Mar 15, 2020 at 7:09
Community Bot
1

answered Mar 26, 2016 at 8:22
EuYu
39.6k 8 85 130

I don't see how those two cycles are identical since $\{b_1, a_n\}$ is used in the second cycle but not in the first. – Mike Daas Mar 26, 2016 at 8:28

@MikeDaas Sorry, you're correct. I labelled the latter cycle incorrectly the first time. Hopefully it's clear this time. – EuYu Mar 26, 2016 at 8:51

I don't want to be too precise, but it might be even clearer if you would remove the b_1 at the beginning of the second cycle, in my opinion – Mike Daas Mar 26, 2016 at 13:02

@MikeDaas Fair enough. – EuYu Mar 27, 2016 at 4:52